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COURSE CODE: CSA 1708

ARTIFICIAL INTELLIGENCE

1.Write the python program to solve 8-Puzzle problem

Python code:

```
from collections import deque


s=input("Enter puzzle: ")

q=deque([(s,[])]) v={s}


while q:

 x,p=q.popleft()  if

x=="123456780":

print("Solution:")

for a in p+[x]:

   print(a[:3]);print(a[3:6]);print(a[6:]);print()

break  i=x.index("0")  for j in [i-3,i+3,i-

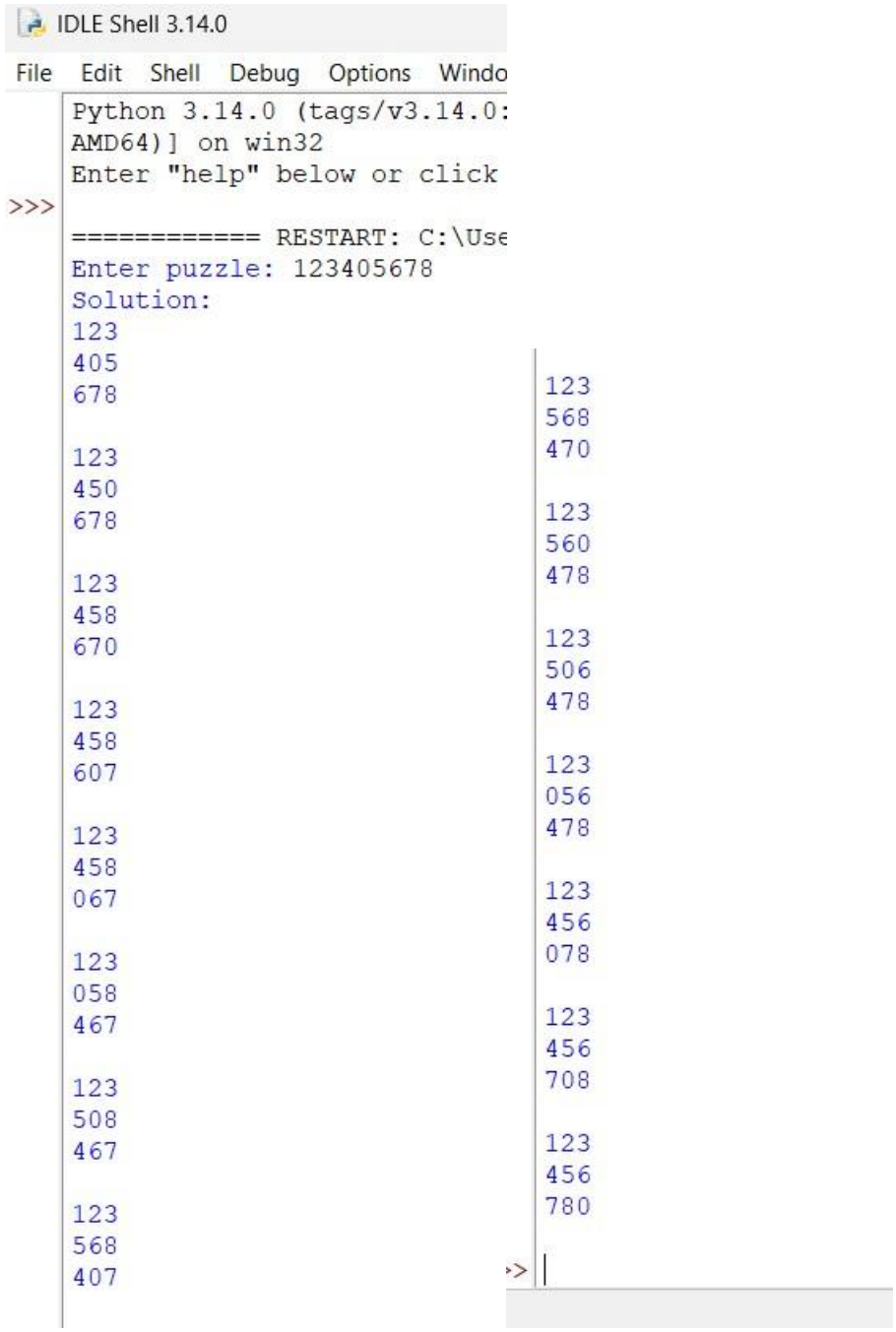
1,i+1]:   if 0<=j<9 and abs(i%3-j%3)<=1:

   a=list(x);a[i],a[j]=a[j],a[i];a=".join(a)

if a not in v:v.add(a);q.append((a,p+[x]))

else:  print("No solution")
```

Output:



2.Write the python program to solve 8-Queen problem

Python code:

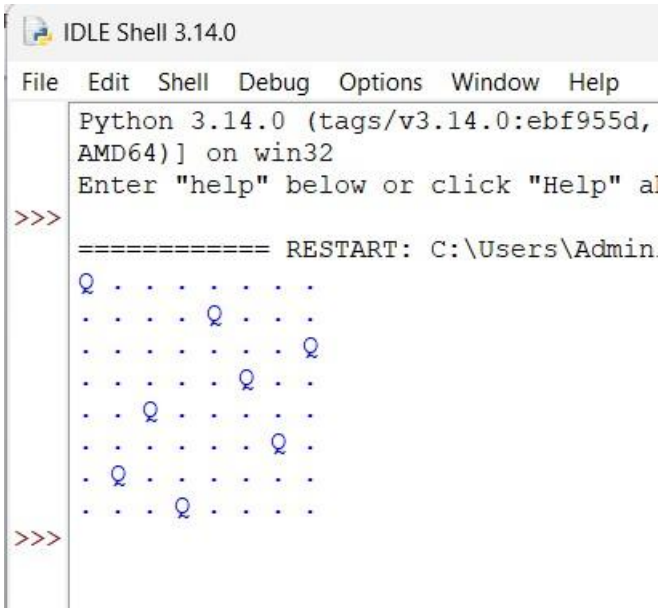
def solve(r, q): if r

== 8: for i in

range(8):

```
            print(" ".join("Q" if q[i] == j else "." for j in range(8)))
return True     for c in range(8):
        if c not in q and all(abs(c-x) != r-i for i,x in enumerate(q)):
            if solve(r+1, q+[c]):
                return True
return False
```

solve(0, []) **Output:**



3.Write the python program for Water Jug Problem

Python code:

a = 0 b =

0 print(a,

b)

while a != 2: if a

== 0: a = 4

elif b == 3: b =

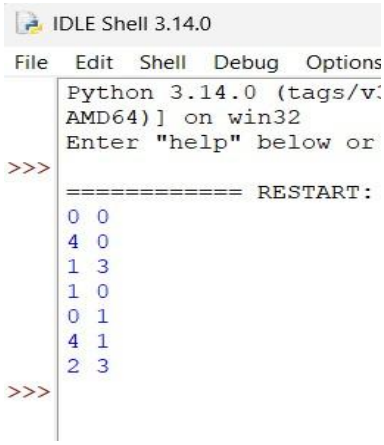
0 else: x =

min(a, 3-b) a =

a-x b = b+x

print(a, b)

Output:



4.Write the python program for Missionaries Cannibal problem

Python code:

from collections import deque

```
q = deque([((3,3,1), [])]) v =

{(3,3,1)} moves =

[(1,0),(2,0),(0,1),(0,2),(1,1)]



while q:

   (m,c,b), p = q.popleft()

if (m,c,b) == (0,0,0):

for x in p + [(m,c,b)]:

          print(x)

break     for dm,dc in

moves:

       nm = m-dm if b else m+dm

nc = c-dc if b else c+dc

       nb = 1-b         if 0<=nm<=3 and 0<=nc<=3:             if

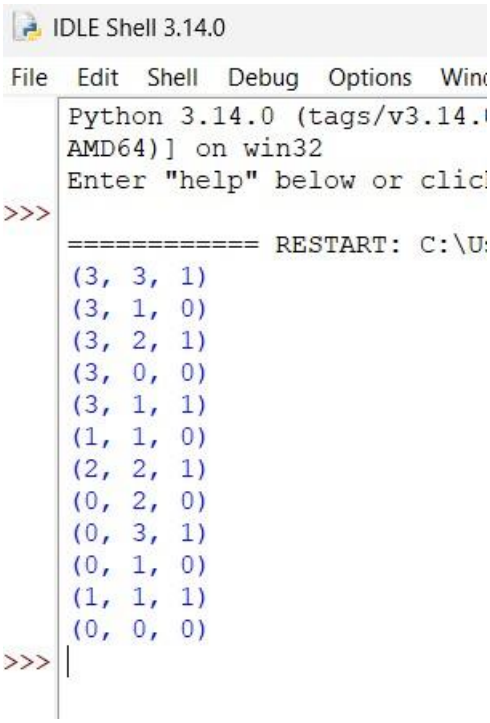
(nm==0 or nm>=nc) and (3-nm==0 or 3-nm>=3-nc):

             s=(nm,nc,nb)

if s not in v:

                v.add(s); q.append((s,p+[(m,c,b)]))
```

Output:



5.Write the python to implement Travelling Salesman Problem

Python code:

```
from itertools import permutations

d=[[0,10,15,20],[10,0,35,25],[15,35,0,30],[20,25,30,0]]

best=999 path=None

for p in permutations(range(1,4)):

   r=(0,)+p+(0,)     cost=sum(d[r[i]][r[i+1]]

for i in range(4))     if cost<best:

best=cost         path=r
```

print("Path:",path) print("Minimum

cost:",best) **Output:**

